OFFLINE STORAGE SYSTEM

Windows Installation & Deployment Guide — v1.0.0

Purpose: Install the packaged application on a Windows laptop or server, run it locally, and optionally allow office users on the same trusted LAN to access it.

1. Download the release

Open GitHub Release v1.0.0 and download **Offline-Storage-Deployment.zip** (approximately 645 MB). The ZIP contains the application, scripts, README, and Windows installers. No separate source-code download is needed.

2. Install prerequisites

Extract the ZIP. Install Node.js first, then MongoDB Community Server using the included installers:

```
installers\nodejs\node-v22.23.5-x64.msi
installers\mongodb\mongodb-windows-x86_64-7.0.48-signed.msi
```

For MongoDB, choose Complete installation and configure it as a Windows Service named **MongoDB** if prompted. Restart Command Prompt after installing Node.js.

Verify installation:

```
node -v npm -v mongosh --version
```

The package was prepared for Node.js 22.23.5 and MongoDB Community Server 7.0.48. Confirm installer compatibility with the target Windows edition and office IT policy before installing on a managed server.

3. Extract the ZIP

Right-click Offline-Storage-Deployment.zip → Extract All. Open the extracted Offline-Storage-Deployment folder. It should contain application, database, installers, scripts, README.txt, and start.bat.

4. Start the application

Double-click **scripts\start.bat**. The script checks Node.js, checks the MongoDB service, attempts to start MongoDB if stopped, and launches the backend. Administrator permission may be needed to start the MongoDB service. Keep the black Command Prompt window open while using the system.

5. Open and verify

On the same computer, open a browser:

```
http://localhost:5000
```

Check backend/database health:

```
http://localhost:5000/api/health
```

A healthy response should contain **"status":"success"** and **"database":"Connected"**. A working health endpoint does not prove the frontend is rendering correctly; verify the dashboard and upload workflows separately.

6. Database behavior

The packaged backend uses application\backend\.env. Default configuration:

```
PORT=5000 MONGODB_URI=mongodb://127.0.0.1:27017/offline_storage
```

MongoDB creates the database when the application first writes data. A new laptop with an empty MongoDB instance will not contain records from the previous computer. Existing records require a separate MongoDB backup and restore process.

7. Office LAN access

The backend listens on 0.0.0.0:5000. To allow other office computers:

- Connect server and client computers to the same trusted office LAN.
- Run ipconfig on the server to find its LAN IPv4 address.
- Ask IT to allow inbound TCP port 5000 in Windows Firewall, limited to the trusted office subnet.
- From another computer, open http://SERVER-IP:5000 (replace SERVER-IP with the server's LAN address).

Security: Keep MongoDB port 27017 restricted to the server itself. Do not expose MongoDB or this application directly to the public internet. Follow office network and security policies.

8. Stop the application

In the Command Prompt window running the backend, press Ctrl+C. MongoDB can remain running. The package's scripts\stop.bat provides stop guidance.

9. Troubleshooting

Problem	Check
'node' is not recognized	Install Node.js, reopen CMD, run node -v.
MongoDB service missing/stopped	Check Windows Services for MongoDB and confirm service installation.
Health says database Disconnected	Confirm MongoDB is running and .env has the correct MONGODB_URI.
Port 5000 already in use	Stop another backend instance (for example, one started from VS Code) before starting deployment.
Dashboard is blank	Check /api/health. If health works, inspect browser errors with F12 → Console; the frontend may have a separate issue.

10. Validation status and release links

The GitHub release ZIP upload was verified. The extracted package's backend health endpoint previously returned success with MongoDB connected. However, the blank dashboard issue and document/audio upload workflows were not fully verified when this guide was prepared. Complete those tests on a clean target computer before production office use.

Source repository
Release v1.0.0 / ZIP download